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Yoshita Srivastava, Ph.D.

Postdoctoral Scholar, RNA Structural Biology

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Summary

Postdoctoral scholar in RNA structural biology and biochemistry, focused on riboswitches, RNA-ligand recognition, and RNA-Fab interfaces.

Education

PhD **University of Rochester Medical Center**, Biophysics — Rochester, NY, USA, 2018 – 2024
- Thesis Advisor: Prof. Joseph E. Wedekind
- Structure and function analysis of bacterial riboswitches that control translation

Integrated BS/MS **Homi Bhabha Research Institute via National Institute of Science Education and Research (NISER)**, Biology — India, 2013 – 2018
- Thesis Advisor: Dr. Rudresh Acharya

Experience

University of Chicago, Post-doctoral Scholar — Chicago, IL, USA, 2024 – present, 2 years

Department of Biochemistry and Molecular Biology, Division of Biological Sciences.
- Characterized binding kinetics and structure-function relationships for RNA-Fab interfaces using BLI and SPR
- Optimized Fab constructs and crystallized an RNA-Fab complex solved to 2.7 Å resolution

University of Rochester Medical Center, Graduate Research Assistant — Rochester, NY, USA, 2018 – 2024, 6 years

Department of Biophysics, Structural, and Computational Biology.
- Crystallized and solved structures of four PreQ1-III riboswitch mutants using Phenix, Coot, XDS, and CCP4
- Optimized a GFP-based bacterial reporter assay for riboswitch mutants
- Characterized RNA secondary structure with SHAPE-seq and analyzed data with Python and GraphPad Prism
- Measured ligand binding thermodynamics and stoichiometry for NAD+-I riboswitches using ITC and SEC
- Developed RNA crystallization and phasing modules by engineering methionine residues and RNA-binding proteins

Homi Bhabha Research Institute, Research Assistant, MS — India, 2016 – 2018, 2 years
- Cloned, expressed, and purified the sensory domain of a Zn-sensing protein (ZraS)

Publications

Two riboswitches that share a common ligand-binding fold show dramatic differences in the ability to accommodate mutations — Nov 2024

Published in 2024.

Srivastava Y., Akinyemi O., Pritchett E.M., Baker C.D., Jenkins J.L., Mathews D.H., Wedekind J.E.

(Nucleic Acids Research)

Full-length NAD+-I riboswitches bind a single co-factor but cannot discriminate against adenosine triphosphate — 2023
Published in 2023.
Srivastava Y., Blau M., Jenkins J.L., Wedekind J.E.
(Biochemistry)

Affinity and structural analysis of the U1A RNA recognition motif with engineered methionines to improve experimental phasing — 2021
Published in 2021.
Srivastava Y., Bonn-Breach R., Chavali S.S., Lippa G.M., Jenkins J.L., Wedekind J.E.
(Crystals)

Translation regulation by the preQ1 riboswitch family — 2023
Published in 2023.
Kiliushik D., Goenner C., Schroeder G.M., Srivastava Y., Jenkins J.L., Wedekind J.E.
(Journal of Biological Chemistry)

Projects

RNA-Fab interface characterization — 2024 – present
Characterized RNA-antibody fragment interfaces by biolayer interferometry and SPR.
- Established structure-function relationships for RNA-Fab binding
- Solved an RNA-Fab complex structure to 2.7 Å resolution

Riboswitch structure and function analysis — 2018 – 2024
Investigated bacterial riboswitch mutants and ligand recognition.
- Solved structures of four PreQ1-III riboswitch mutants
- Characterized NAD+-I riboswitch ligand recognition and thermodynamics

Volunteer

American Crystallographic Association, Volunteer, technical support — Virtual
- Supported sessions for the annual conference — 2021 – 2021

Department of Biochemistry & Biophysics interview weekend, Volunteer — University of Rochester
- Served as host and presented a research poster — 2022 – 2022

Alliance for Diversity in Science and Engineering, Communications Team Member — University of Rochester
2021 – 2022

Awards

Muttaiya Sundaralingam/Linus Pauling Poster Award — Aug 2022
Portland, OR.
American Crystallographic Association Annual Meeting

Student Travel Award — 2022
Portland, OR.
American Crystallographic Association

Neuman Travel Award — 2020
Spring 2020.
Department of Biochemistry and Biophysics, University of Rochester

KVPY Fellowship — 2014
Renewed yearly through 2018.
Department of Science and Technology, Government of India

Talks and Presentations

Chemistry-Biology Interface Retreat, University of Rochester, Talk — Rochester, NY, USA, 2023 – 2023
Two riboswitches that share a common ligand-binding site show dramatic differences in their ability to accommodate mutations.

American Crystallographic Association Annual Meeting, Poster — Portland, OR, USA, 2022 – 2022
Two riboswitches that share a common ligand-binding fold show dramatic differences in the ability to accommodate mutations.

Microbes and RNA Conference, Poster — Virtual, 2022 – 2022
Analysis of cofactor binding by NAD riboswitches.

Workshops and Seminars

Mentor's training workshop for postdocs, Workshop — Virtual, 2025 – 2025

NE-CAT data collection workshop, Argonne National Laboratory, Workshop — Argonne, IL, USA, 2025 – 2025

Hands-on Single-Particle Cryo-EM Data Analysis with cryoEDU, Seminar — American Crystallographic Association Annual Meeting, 2022 – 2022

Teaching and Mentoring Experience

University of Rochester, Teaching Assistant, IND408 Advanced Biochemistry — Rochester, NY, USA, 2019 – 2019
- Supported graduate and senior undergraduate instruction

University of Rochester, Project Supervisor — Rochester, NY, USA, 2022 – 2022
- Supervised undergraduate research coursework

University of Rochester, Lab Rotation Supervisor — Rochester, NY, USA, 2021 – 2022
- Supervised three graduate student rotations

Skills

Structural Biology

Biochemical Methods

Computational

Languages

English
Fluent

Hindi
Fluent

Community Outreach

American Crystallographic Association, Volunteer, technical support — Virtual, 2021 – 2021
- Supported conference sessions

Department of Biochemistry & Biophysics interview weekend, Volunteer — University of Rochester, 2022 – 2022
- Hosted visitors and presented research

Alliance for Diversity in Science and Engineering, Communications Team Member University of Rochester
2021 – 2022